THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 11: **Earth, Moon and the Sun** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** have continuous daylight	No change in facing.
2	**(b)** evening sky, in the west	Soon after sunset.
3	**(d)** umbra	Total.
4	**(a)** no seasons	Same sunlight all year.
5	**(b)** sunset	Opposite the Sun.
6	**(c)** penumbra	Partial shadow.
7	**(a)** northern	North tilted to Sun.
8	**(b)** sunrise	Along with the Sun.
9	**(c)** the Earth's shadow is large	Larger shadow.
10	**(c)** summer	Opposite.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Axis is tilted (1); Sun above horizon for different durations (1).	1+1
12	No air or water (1); No weathering (1).	1+1
13	Rays slant and spread (1); Shorter days (1).	1+1
14	Sun's rays are always nearly direct (1); Day length is nearly constant (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	New moon aligned with Sun and Earth (1); Moon's umbra falls on Earth (1); Region in umbra sees total eclipse (1).	1+1+1
16	Near the poles (1); Sun above horizon at midnight (1); During summer (1).	1+1+1
17	Lower gravity (1); One-sixth (1); Easier (1).	1+1+1
	OR: Lamp as Sun (1); Ball moved around (1); Lit part visible changes (1).	1+1+1
18	Kharif in rains (1); Rabi in winter (1); Temperature and water matter (1).	1+1+1
	OR: The Moon blocks the Sun's light (1); Umbra falls on the place (1); Day looks like dusk (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	21 June: north summer (1); 22 Dec: north winter (1); Equinoxes: equal day and night (1); South opposite (1); Tilt same direction (1).	1+1+1+1+1
	OR: The Moon turns once on its axis in about 27 days (1); It also goes round the Earth in about 27 days (1); So the same face points towards the Earth (1); The far side is never seen from the Earth (1); The far side still gets sunlight (1).	1+1+1+1+1
20	New moon (1); Waxing crescent, first quarter (1); Waxing gibbous, full moon (1); Waning gibbous, last quarter (1); Waning crescent (1).	1+1+1+1+1
	OR: Night light (1); Lunar calendar (1); Festivals (1); Moon missions (1); Phases teach geometry of light (1).	1+1+1+1+1

Mistakes to avoid:

- Saying that seasons are caused by the Earth being closer to or farther from the Sun; they are caused by the **tilt of the axis** and revolution.
- Mixing up rotation (day and night, 24 hours) with revolution (year, $365\frac{1}{4}$ days).
- Interchanging eclipses: **solar** eclipse = Moon between Sun and Earth (new moon); **lunar** eclipse = Earth between Sun and Moon (full moon).
- Writing that the Moon gives its own light; it only **reflects** sunlight.
- Thinking the phases of the Moon are caused by the Earth's shadow; they are due to how much of the sunlit half we see.
- Writing that it is summer in both hemispheres at the same time; the seasons are **opposite** in the two hemispheres.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 11: **Earth, Moon and the Sun** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** half disc, right half lit (northern hemisphere)	Quarter of the way.
2	**(c)** dark reddish	Sunlight bends through atmosphere.
3	**(a)** 12 hours	Equal.
4	**(c)** the Sun always shines on half of it	One side faces Sun.
5	**(b)** lunar	Only lunar.
6	**(a)** the Sun does not set	Polar day.
7	**(c)** 15	Half of the cycle.
8	**(b)** in a narrow region	Small shadow.
9	**(d)** the Sun does not rise	Polar night.
10	**(b)** 7	A quarter of the cycle.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Time between two full moons (1); About 29.5 days (1).	1+1
12	Axis tilt (1); Sun above or below horizon (1).	1+1
13	Moon revolves around Earth (1); Moves eastward in its orbit (1).	1+1
14	Narrow shadow (1); Fewer places see it (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	June longer (1); December shorter (1); Axis tilt (1).	1+1+1
16	Reflects only part of sunlight (1); Dark surface (1); Far away (1).	1+1+1
17	Higher in summer (1); Lower in winter (1); Due to tilt (1).	1+1+1
	OR: Tilt gives variation (1); Tropics have less change (1); India has monsoon seasons (1).	1+1+1
18	Moon moves (1); Shadow moves (1); Small shadow (1).	1+1+1
	OR: Sun highest (1); Rays most vertical (1); Shortest shadow (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Solar: Moon between (1); Lunar: Earth between (1); Alignment needed (1); Moon's orbit tilted (1); Rare (1).	1+1+1+1+1
	OR: Observed shadows (1); Noted phases (1); Eclipses predicted (1); Panchang (1); Aryabhata (1).	1+1+1+1+1
20	Lamp as Sun (1); Tilted globe (1); Move around keeping the tilt (1); Observe sunlight on each hemisphere (1); Seasons seen (1).	1+1+1+1+1
	OR: Make a small hole in a card (1); Hold it with its shadow on paper (1); See crescent image (1); Do not look at the Sun (1); Adults supervise (1).	1+1+1+1+1

Mistakes to avoid:

- Saying that seasons are caused by the Earth being closer to or farther from the Sun; they are caused by the **tilt of the axis** and revolution.
- Mixing up rotation (day and night, 24 hours) with revolution (year, $365\frac{1}{4}$ days).
- Interchanging eclipses: **solar** eclipse = Moon between Sun and Earth (new moon); **lunar** eclipse = Earth between Sun and Moon (full moon).
- Writing that the Moon gives its own light; it only **reflects** sunlight.
- Thinking the phases of the Moon are caused by the Earth's shadow; they are due to how much of the sunlit half we see.
- Writing that it is summer in both hemispheres at the same time; the seasons are **opposite** in the two hemispheres.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 11: **Earth, Moon and the Sun** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** wherever the Moon is above the horizon	Night side.
2	**(c)** 21 June	Summer solstice.
3	**(d)** 29.5 days	Synodic month.
4	**(a)** the Moon's orbit is tilted	Alignment not exact.
5	**(c)** 22 December	Winter solstice.
6	**(d)** 27 days	Same as revolution.
7	**(b)** dark in the day	Sun blocked.
8	**(d)** equinoxes	21 March and 23 September.
9	**(b)** its mass is less	Smaller mass.
10	**(a)** sudden darkness	They respond to light.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Moon reflects it (1); Without it, no phases (1).	1+1
12	Moon's shadow is small (1); Falls on a narrow strip (1).	1+1
13	To match the year with Earth's revolution (1); Quarter-day adjustment (1).	1+1
14	It lies in the direction of Earth's axis (1); Earth spins about this line (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Moon is above the horizon (1); Sun not too bright to see it (1); Orbit causes it (1).	1+1+1
16	Earth's atmosphere bends red light (1); Reaches the Moon (1); Blue is scattered (1).	1+1+1
17	Everything moves together (1); Speed is steady (1); No sensation (1).	1+1+1
	OR: The southern hemisphere is tilted towards the Sun (1); Rays are more direct and days are longer (1); Distance from the Sun is not the cause (1).	1+1+1
18	Months based on phases (1); Purnima, Amavasya (1); Festivals fixed by it (1).	1+1+1
	OR: No seasons (1); Equal day and night (1); Poles always cold (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Day from rotation (1); Year from revolution (1); Seasons for crops (1); Calendars (1); Leap year (1).	1+1+1+1+1
	OR: Longest around 21 June (1); Shortest around 22 December (1); Equal at equinox (1); Gradual change (1); Tilt effect (1).	1+1+1+1+1
20	Higher latitude has more variation (1); Srinagar longer in June (1); Shorter in December (1); Mumbai close to equator (1); Tilt (1).	1+1+1+1+1
	OR: Moon's umbra is narrow (1); Falls on a thin strip (1); Alignment is rare (1); Lasts only minutes (1); Travel to the path to see it (1).	1+1+1+1+1

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